

```
pip install pandas numpy matplotlib seaborn scikit-learn nltk transformers torch
```

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Requirement already satisfied: pandas in /usr/local/lib/python3.12/dist-packages (2.2.2)
Requirement already satisfied: numpy in /usr/local/lib/python3.12/dist-packages (2.0.2)
Requirement already satisfied: matplotlib in /usr/local/lib/python3.12/dist-packages (3.10.0)
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Requirement already satisfied: scikit-learn in /usr/local/lib/python3.12/dist-packages (1.6.1)
Requirement already satisfied: nltk in /usr/local/lib/python3.12/dist-packages (3.9.1)
Requirement already satisfied: transformers in /usr/local/lib/python3.12/dist-packages (5.0.0)
Requirement already satisfied: torch in /usr/local/lib/python3.12/dist-packages (2.10.0+cpu)
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Requirement already satisfied: certifi in /usr/local/lib/python3.12/dist-packages (from httpx<1,>=0.23.0->huggingface-hub<2.0,>=1.3.0) (2025.11.3)
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Requirement already satisfied: annotated-doc>=0.0.2 in /usr/local/lib/python3.12/dist-packages (from typer->huggingface-hub<2.0,>=1.3.0) (0.0.2)
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```

```
import pandas as pd
import numpy as np
import matplotlib.pyplot as plt
import seaborn as sns

from sklearn.model_selection import train_test_split
from sklearn.feature_extraction.text import TfidfVectorizer
from sklearn.linear_model import LogisticRegression
from sklearn.metrics import accuracy_score
from sklearn.metrics import classification_report
from sklearn.metrics import confusion_matrix
from sklearn.metrics import ConfusionMatrixDisplay

import nltk
nltk.download('stopwords')
nltk.download('punkt')

from nltk.corpus import stopwords
from nltk.tokenize import word_tokenize
```

```
[nltk_data] Downloading package stopwords to /root/nltk_data...
[nltk_data] Unzipping corpora/stopwords.zip.
```

```
[nltk_data] Downloading package punkt to /root/nltk_data...
[nltk_data] Unzipping tokenizers/punkt.zip.
```

```
df = pd.read_csv("/content/internship_feedback_sentiment.csv")
```

```
df.head()
```

	feedback_id	feedback_text	sentiment_label
0	FB_00001	This internship exceeded my expectations. The ...	positive
1	FB_00002	My mentor was great teacher. The Business Anal...	positive
2	FB_00003	The internship experience was highly beneficia...	positive
3	FB_00004	This internship exceeded my expectations. The ...	positive
4	FB_00005	This internship exceeded my expectations. The ...	positive

```
df.shape
```

```
(2000, 3)
```

```
df.info()
```

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 2000 entries, 0 to 1999
Data columns (total 3 columns):
#   Column          Non-Null Count  Dtype
---  ---
0   feedback_id      2000 non-null   object
1   feedback_text     2000 non-null   object
2   sentiment_label   2000 non-null   object
dtypes: object(3)
memory usage: 47.0+ KB
```

```
df.describe()
```

	feedback_id	feedback_text	sentiment_label
count	2000	2000	2000
unique	2000	420	3
top	FB_01984	This internship exceeded my expectations. The ...	positive
freq	1	65	1305

```
print(df.isnull().sum())
```

```
feedback_id      0
feedback_text     0
sentiment_label   0
dtype: int64
```

```
df.dropna(inplace=True)
```

```
df = df.drop("feedback_id", axis =1)
```

```
stop_words = set(stopwords.words('english'))
```

```
def clean_text(text):

    words = word_tokenize(text.lower())

    filtered_words = []

    for word in words:
        if word.isalnum() and word not in stop_words:
            filtered_words.append(word)

    return " ".join(filtered_words)
```

```
import nltk
nltk.download('punkt_tab')
df['Cleaned_Feedback'] = df['feedback_text'].apply(clean_text)
```

```
[nltk_data] Downloading package punkt_tab to /root/nltk_data...
[nltk_data] Unzipping tokenizers/punkt_tab.zip.
```

```
vectorizer = TfidfVectorizer()

X = vectorizer.fit_transform(df['Cleaned_Feedback'])
```

```
y = df['sentiment_label']
```

```
X_train, X_test, y_train, y_test = train_test_split(
    X,
    y,
    test_size=0.2,
    random_state=42
)
```

```
model = LogisticRegression()

model.fit(X_train, y_train)
```

```
▼ LogisticRegression ⓘ ?
LogisticRegression()
```

```
predictions = model.predict(X_test)
```

```
accuracy = accuracy_score(y_test, predictions)

print("Accuracy:", accuracy)
```

```
Accuracy: 1.0
```

```
print(classification_report(y_test, predictions))
```

	precision	recall	f1-score	support
negative	1.00	1.00	1.00	43
neutral	1.00	1.00	1.00	93
positive	1.00	1.00	1.00	264
accuracy			1.00	400
macro avg	1.00	1.00	1.00	400
weighted avg	1.00	1.00	1.00	400

```
cm = confusion_matrix(y_test, predictions)
```

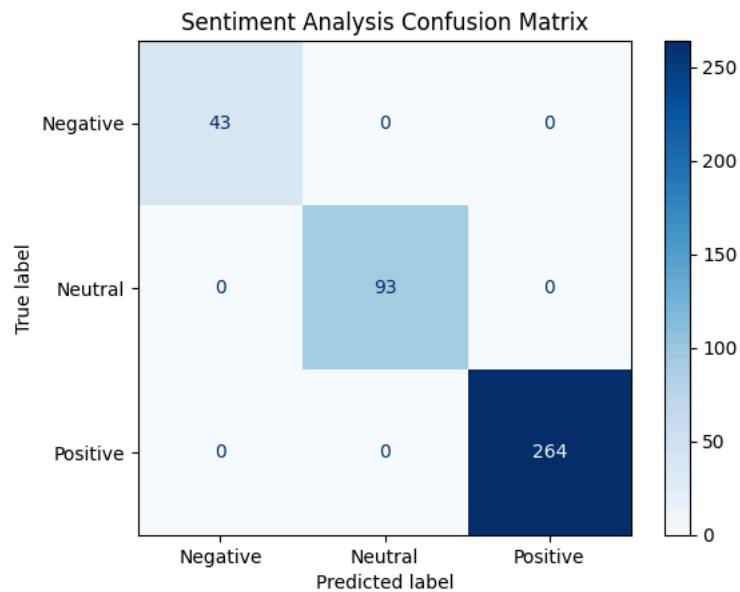
```
labels = ['Negative', 'Neutral', 'Positive']

cmd = ConfusionMatrixDisplay(
    confusion_matrix=cm,
    display_labels=labels
)

cmd.plot(cmap='Blues')

plt.title("Sentiment Analysis Confusion Matrix")

plt.show()
```



```
new_feedback = ["The internship experience was very helpful and amazing"]
```

```
cleaned_feedback = []
```

```
for text in new_feedback:
    cleaned_feedback.append(clean_text(text))
```

```
vectorized_feedback = vectorizer.transform(cleaned_feedback)
```

```
prediction = model.predict(vectorized_feedback)
```

```
print("Predicted Sentiment:", prediction[0])
```

```
Predicted Sentiment: positive
```

```
feature_names = vectorizer.get_feature_names_out()
```

```
coefficients = model.coef_[0]
```

```
top_features = pd.DataFrame({
    'Word': feature_names,
    'Importance': coefficients
})
```

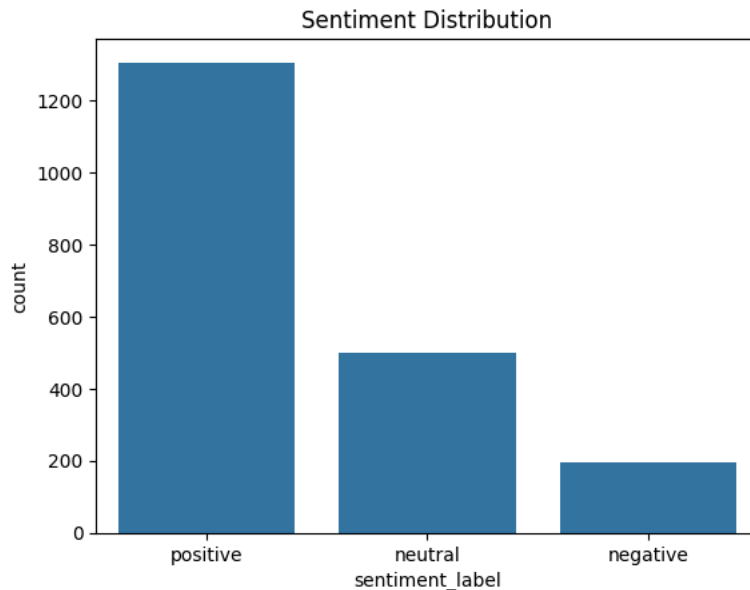
```
print(top_features.sort_values(by='Importance', ascending=False).head(10))
```

	Word	Importance
89	needs	1.896974
3	affected	1.753360
116	significantly	1.753360
138	unfortunately	1.485000
50	faced	1.485000
20	challenges	1.485000
34	difficult	1.474031
16	benefit	1.417657
56	fully	1.417657
35	disappointing	1.367256

```
sns.countplot(x='sentiment_label', data=df)
```

```
plt.title("Sentiment Distribution")
```

```
plt.show()
```



```
pip install transformers
```

```
Requirement already satisfied: transformers in /usr/local/lib/python3.12/dist-packages (5.0.0)
Requirement already satisfied: filelock in /usr/local/lib/python3.12/dist-packages (from transformers) (3.29.0)
Requirement already satisfied: huggingface-hub<2.0,>=1.3.0 in /usr/local/lib/python3.12/dist-packages (from transformers) (1.11.0)
Requirement already satisfied: numpy>=1.17 in /usr/local/lib/python3.12/dist-packages (from transformers) (2.0.2)
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Requirement already satisfied: tqdm>=4.27 in /usr/local/lib/python3.12/dist-packages (from transformers) (4.67.3)
Requirement already satisfied: fsspec>=2023.5.0 in /usr/local/lib/python3.12/dist-packages (from huggingface-hub<2.0,>=1.3.0->transformers) (2025.10.0)
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Requirement already satisfied: h11>=0.16 in /usr/local/lib/python3.12/dist-packages (from httpcore==1.*->httpx<1,>=0.23.0->huggingface-hub<2.0,>=1.3.0->transformers) (0.14.0)
Requirement already satisfied: click>=8.2.1 in /usr/local/lib/python3.12/dist-packages (from typer->huggingface-hub<2.0,>=1.3.0->transformers) (8.1.8)
Requirement already satisfied: shellingham>=1.3.0 in /usr/local/lib/python3.12/dist-packages (from typer->huggingface-hub<2.0,>=1.3.0->transformers) (1.5.4)
Requirement already satisfied: rich>=12.3.0 in /usr/local/lib/python3.12/dist-packages (from typer->huggingface-hub<2.0,>=1.3.0->transformers) (13.9.0)
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```

```
from transformers import pipeline

classifier = pipeline("sentiment-analysis")

result = classifier("The internship was excellent")

print(result)
```

```
No model was supplied, defaulted to distilbert/distilbert-base-uncased-finetuned-sst-2-english and revision 714eb0f.
Using a pipeline without specifying a model name and revision in production is not recommended.
/usr/local/lib/python3.12/dist-packages/huggingface_hub/utils/_auth.py:93: UserWarning:
The secret `HF_TOKEN` does not exist in your Colab secrets.
To authenticate with the Hugging Face Hub, create a token in your settings tab (https://huggingface.co/settings/tokens), set it
You will be able to reuse this secret in all of your notebooks.
Please note that authentication is recommended but still optional to access public models or datasets.
warnings.warn(
Warning: You are sending unauthenticated requests to the HF Hub. Please set a HF_TOKEN to enable higher rate limits and faster d
WARNING:huggingface_hub.utils._http:Warning: You are sending unauthenticated requests to the HF Hub. Please set a HF_TOKEN to en
config.json: 100%                               629/629 [00:00<00:00, 17.5kB/s]
model.safetensors: 100%                         268M/268M [00:02<00:00, 211MB/s]
```